

Multiple Choice

1. **Answer:** C

Options: A) Docosahexaenoic acid can be synthesised from linolenic acid in vegans; B) Docosahexaenoic acid is absent from vegan diets; C) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment; D) Microalgae can be used to synthesise docosahexaenoic acid

Note: The statement is not true because while docosahexaenoic acid (DHA) is important for brain and eye health, vegans can obtain adequate amounts of DHA through algae-based supplements or by converting alpha-linolenic acid (ALA) from plant sources into DHA.

2. **Answer:** A

Options: A) <10% of total daily energy; B) <12%; C) <15%; D) <16%

Note: The recommendation for individuals with diabetes to limit saturated fat to less than 10% of total daily energy is based on evidence that reducing saturated fat intake can help improve insulin sensitivity and lower the risk of cardiovascular complications associated with diabetes.

3. **Answer:** C

Options: A) Increased bone mass; B) Increased muscle mass; C) Increased body fat; D) All of the above

Note: In older adults, hormonal changes, decreased physical activity, and loss of muscle mass typically lead to an increase in body fat as part of the natural ageing process.

4. **Answer:** D

Options: A) No account is taken of waste or losses of food; B) Consumption estimates give higher values than those derived from other types of surveys; C) No information is available on the distribution of consumption within a country; D) All of the options given are correct

Note: The correct answer is that all of the options given are correct because food balance sheets often have limitations such as incomplete data, potential inaccuracies in reporting, and failure to account for food waste, which collectively hinder their effectiveness in providing a comprehensive view of food availability and consumption.

5. **Answer:** D

Options: A) Perfectionism traits; B) Female gender; C) Dieting during adolescence; D) Parental anxiety

Note: Parental anxiety is not considered a direct risk factor for eating disorders, as research primarily links eating disorders to factors such as genetics, environmental influences, and sociocultural pressures rather than specifically to the emotional states of parents.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Epigenetic changes can lead to different phenotypes for the same genotype due to variations in environmental factors and individual experiences that influence gene expression.

7. **Answer:** False

Options: A) True; B) False

Note: Proline and serine are classified as non-essential amino acids because the body can synthesize them, while tyrosine is considered conditionally essential as it can be produced from phenylalanine when dietary intake is sufficient.

8. **Answer:** False

Options: A) True; B) False

Note: Protein-energy supplementation can benefit individuals of all ages, including pregnant women and young children, by supporting growth and development during critical periods of life.

9. **Answer:** True

Options: A) True; B) False

Note: Folate, vitamin B₆, and vitamin B₁₂ are essential nutrients that help convert homocysteine into other substances in the body, thereby reducing its levels in the bloodstream.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the World Health Organization (WHO) advises that reducing free sugar intake to below 10% of total energy consumption can decrease the risk of dental caries, obesity, and other health issues.

Long Form Response

11. **Answer:** Infants suffering from severe combined immunodeficiency (SCID) may exhibit signs such as recurrent infections, failure to thrive, and chronic diarrhea. They may also show unusual skin rashes or persistent oral thrush. These indicators are crucial for early diagnosis and treatment because SCID significantly weakens the immune system, making infants highly vulnerable to infections that can be life-threatening. Early detection allows for timely interventions, such as bone marrow transplants or gene therapy, which can greatly improve the child's health and survival chances. Recognizing these symptoms promptly is essential for providing the necessary care and support to affected infants.

Note: The answer correctly identifies critical signs of severe combined immunodeficiency, such as recurrent infections and failure to thrive, and emphasizes the impor-

tance of early diagnosis for effective treatment, highlighting the urgent need to address the life-threatening vulnerabilities of affected infants.

12. **Answer:** Carbohydrates play a significant role in oral health, as they can influence the growth of bacteria in the mouth. Some carbohydrates, like sugars, can be easily metabolized by oral bacteria, leading to the production of acids that contribute to tooth decay. In contrast, starch, which is a complex carbohydrate, cannot be metabolized by oral bacteria in the same way because the bacteria lack the necessary enzymes to break it down. This means that starch does not promote the same harmful acid production, making it less detrimental to dental health. Therefore, while all carbohydrates can impact oral health, the type of carbohydrate matters significantly in determining its effects.

Note: correct because it accurately distinguishes between simple carbohydrates, which are readily metabolized by oral bacteria to produce harmful acids, and starch, a complex carbohydrate that oral bacteria cannot effectively break down due to the absence of necessary enzymes, thereby reducing its potential to contribute to tooth decay.